

Recent Advances Technologies in Financial Time-Series Generation: A Survey

Abstract

The substantial capacity of large models (LMs) to process long sequences has sparked growing interest in time-series analysis, especially temporal forecasting. In the finance domain, emerging agentic systems driven by LM stand as powerful tools for accurately forecasting future trends in investment products, significantly enhancing investors' decision-making. Considerable effort has thus been dedicated to meeting the demands of these models for extensive data resources. In this paper, we conduct a systematic review of advanced technologies for financial time series data generation (FTSDG), aiming to provide a comprehensive overview of this rapidly evolving field. We categorize FTSDG into three primary topics based on specific applications, including sequence generation, interval imputation, and data augmentation. For each topic, we present a technical framework that covers the standpoints from traditional statistical learning methods to contemporary large diffusion and language models, accompanied by comprehensive comparisons. Additionally, we compile a collection of high-quality FTS datasets, detailing their subjects, volumes, accessibility, and relevant attributes to facilitate utilization. Our project is maintained on GitHub as a persistent contribution to the community.

1 Introduction

Time-series analysis (TSA) is crucial in the financial context, which primarily involves modeling prices, returns, trading volumes, and other market factors. It enables the understanding of marketing dynamics, trend forecasting, risk management, and ultimately supports rational decision-making. As the de facto tools of TSA evolve from classical temporal-frequency analysis to deep neural networks, the growing demand for real-world data clashes with privacy concerns. This challenge ignited interest in a rising field: the study of financial time-series data generation (FTSDG). Recent advancements in generative artificial intelligence, including ChatGPT, StableDiffusion, and DeepSeek, inspired the harnessing of large-scale models to generate financial time series, leading to significant milestones.

Existing literature has explored key technologies for time-series generation. Kong et al. [Kong *et al.*, 2025] reviewed deep learning (DL) methods, particularly emphasizing model development and feature extraction. Kim et al. [Kim *et al.*, 2025] conducted a similar work, going deeper into the specific problems each method addressed. To align with contemporary technical trends, Zhang et al. [Zhang *et al.*, 2024] and Liang et al. [Liang *et al.*, 2024] reviewed LLMs specifically tailored for time-series generation, respectively. As for FTSDG, Islam et al. [Islam *et al.*, 2024] and Alonso et al. [Alonso *et al.*, 2024] conducted case studies, comparing emerging financial LLMs (with traditional DL models) and demonstrating their potential in financial time-series forecasting. Nie et al. [Nie *et al.*, 2024] comprehensively review financial LLMs designed for contextual applications. Arsenault et al. [Arsenault *et al.*, 2025] offer a unique and detailed exploration of explainable financial time-series forecasting models within the machine learning (ML) context. Despite their inspirations, these works present a constrained perspective on tasks, as time-series forecasting is not all that time-series data generation indicates. Their technological focus is also narrowed to LLMs or classical ML, neglecting contemporary advancements such as diffusion models. Therefore, a brand new survey of FTSDG, encompassing a broader scope of tasks and an updated tech map, is required.

In this paper, we comprehensively review over 100 articles published in the last three years and take a snapshot of the cutting-edge technologies of FTSDG. As depicted in Figure 1, we categorize FTSDG into three distinct topics depending on their applications: sequence generation (FTSG), interval imputation (FTSI), and data augmentation (FTSDA). These three topics correspond to situations with available time-series sequences in short form, fragmentary form, and within a limited volume, respectively. FTSG generates subsequent time series conditioned on a given starting span, FTSI predicts intervals missed in incomplete time series, and FTSA enhances the diversity and volume of a time series dataset, similar to open-domain data augmentation. To our knowledge, this taxonomy recognizes almost all current time-series generation approaches. Additionally, we outline various technical schools associated with each topic, ranging from statistical machine learning to deep learning, as well as modern pre-trained large models. We underscore the technical diversity of FTSDG, in contrast to the monotonous paradigms seen in

visual and language generations. To promote the effective use of relevant data resources, we also gather existing datasets for FTSDG, detailing their attributes such as tasks, volume, and accessibility. The remainder is structured as follows: FTSDG approaches are reviewed by topic in Sections 2, 3, and 4, and relevant datasets are compiled in Section 5. Promising future directions are outlined in Section 6 as a conclusion.

2 Financial time-series forecasting

2.1 Task Definition

Let $\Omega = \{y_1, y_2, \dots, y_T\}$ denote an observed financial time series of length T , where each $y_t \in \mathbb{R}^d$ contains d contemporaneous variables (prices, returns, order-flow, etc.). The sample path Ω is assumed to be drawn from an unknown joint probability measure $P(\Omega)$ defined on the measurable space $(\Omega, \mathcal{B})$ of infinite-length trajectories. Owing to market microstructure noise, regime shifts, and extreme events, $P(\Omega)$ is non-stationary, heavy-tailed, and exhibits strong temporal and cross-sectional dependence. Forecasting models are required to estimate a conditional distribution $P(y_{T+1:T+H} | \Omega)$ for horizon H , but only a single historical realisation Ω is available. The scarcity of data, combined with the low probability of tail episodes, leads to high-variance estimators and poor generalisation.

Financial Time Series Data Augmentation (FSTDA) is therefore formalised as the task of constructing a controlled synthesis operator

$$G_\theta : \mathcal{Z} \times \mathcal{C} \rightarrow \hat{\Omega}, \quad (z, c) \mapsto \hat{\Omega}$$

where $\mathcal{Z}$ is a latent noise space, $\mathcal{C}$ is a user-specified conditioning set (macro scenarios, volatility targets, regulatory shocks), and θ parameterises the generator. The synthetic set $\hat{\Omega} = \{\hat{y}_1, \hat{y}_2, \dots, \hat{y}_T\}$ must satisfy:

1. **Distributional fidelity:** $D(P(\hat{\Omega}), P(\Omega)) \leq \varepsilon_1$ under a chosen probability metric D (e.g., Wasserstein, MMD, KL);
2. **Temporal coherence:** $\rho(\hat{\Omega}) \approx \rho(\Omega)$ where $\rho(\cdot)$ denotes stylised facts (autocorrelation, volatility clustering, leverage effect, extreme co-movement);
3. **Conditional alignment:** $\mathbb{E}[\varphi(\hat{\Omega}) | c] = c \pm \varepsilon_2$ for any macro-moment or risk-factor functional φ (return, realised vol, draw-down, correlation breakdown);
4. **Privacy guarantee:** the release of $\hat{\Omega}$ is (ε, δ) -differentially private with respect to the underlying individual records when micro-level data are used.

The augmented corpus $\Omega^* = \Omega \cup \hat{\Omega}$ (or re-weighted mixtures thereof) is then fed to downstream forecasting or decision models to reduce estimation variance, improve tail coverage, and enable stress-testing without multiplying real-market observations.

2.2 Metrics

Mean Squared Error (MSE). MSE gauges the average squared distance between each predicted value and the realised value, thereby magnifying the influence of large errors.

It is computed as

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where n is the number of forecasts. By penalising outliers quadratically, MSE provides a single scalar that summarises both the bias and the variance of a point predictor; traders and risk managers use it when rare but extreme misses must be explicitly discouraged.

Root Mean Squared Error (RMSE). RMSE restores the original units by taking the square root of MSE, yielding an interpretable “average dollar (or basis-point) mistake”. The formula reads

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}.$$

Because it is expressed in the same scale as the target variable, RMSE is often preferred when communicating forecast accuracy to stakeholders who think in price or yield levels rather than squared deviations.

Mean Absolute Error (MAE). MAE averages the absolute discrepancies, offering a linear penalty that is less sensitive to extreme errors than MSE. Formally,

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|.$$

Its robustness to fat-tailed shocks makes MAE attractive for financial series where sporadic jumps are part of the data-generating process and should not dominate the performance picture.

Mean Absolute Percentage Error (MAPE). MAPE rescales each error by the realised value and reports the average percentage deviation,

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|.$$

By expressing mistakes in relative terms, MAPE allows comparison across assets with different price levels or volatilities and is routinely cited in equity- and FX-return forecasts where basis-point accuracy matters.

Information Coefficient (IC). IC measures the monotonic relationship between forecast and outcome ranks, typically via Spearman’s ρ ,

$$\text{IC} = \rho(\text{rank}(\hat{y}), \text{rank}(y)).$$

A high IC signifies that the model correctly orders securities or time periods, which is exactly what ranking-based portfolios (e.g., quantile long-short strategies) require; hence IC is often the primary KPI in buy-side research.

Directional Accuracy (Hit Ratio). This metric counts how often the sign of the forecast matches the sign of the realisation,

$$\text{Accuracy} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{sign}(\hat{y}_i) = \text{sign}(y_i)].$$

For market-timing or binary up/down decisions, directional accuracy translates directly into the probability of being on the right side of the move, making it intuitive for traders even when point-error magnitudes are small.

Continuous Ranked Probability Score (CRPS). CRPS integrates the squared difference between the forecast CDF F and the empirical step function of the observation,

$$\text{CRPS} = \int [F(z) - \mathbf{1}\{z \geq y\}]^2 dz.$$

It rewards both calibration (probabilities match frequencies) and sharpness (narrow distributions), providing a single score for density, quantile, or ensemble forecasts that collapses to MAE when only a point estimate is supplied.

Quantile Score (QS or Pinball Loss). For a target quantile τ , QS imposes an asymmetric linear penalty,

$$\text{QS}_\tau = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{q}_{\tau,i}) (\tau - \mathbf{1}\{y_i \leq \hat{q}_{\tau,i}\}).$$

Minimising QS ensures that the τ -th quantile of the predictive distribution is hit with the correct frequency, which is essential for Value-at-Risk or expected-shortfall models used in regulatory and risk-management frameworks.

Sharpe Ratio. The Sharpe ratio divides the excess return generated by a forecast-based strategy over the risk-free rate by its standard deviation,

$$\text{Sharpe} = \frac{\mathbb{E}(R_p) - R_f}{\sigma(R_p)}.$$

It converts raw performance into a risk-adjusted reward measure, allowing investors to compare strategies across different asset classes or leverage levels on a common unit-less scale.

Calmar Ratio. Calmar replaces volatility with maximum drawdown,

$$\text{Calmar} = \frac{\mathbb{E}(R_p) - R_f}{\text{MaxDD}}.$$

By focusing on the worst peak-to-trough loss rather than second-moment dispersion, it appeals to mandate-constrained or leveraged funds whose survival depends on limiting tail downside rather than on controlling ordinary volatility.

2.3 Works Review

Financial time-series forecasting seeks to infer future trajectories of asset prices, returns, or risk indicators from their historical realisations together with related macro- and micro-structural information, thereby furnishing an evidential basis for portfolio allocation, risk hedging, and regulatory decision-making. Empirical datasets in finance are, however, constrained by privacy statutes, compliance mandates, and prohibitive collection costs; publicly available samples are typically limited in length, dimensionality, and diversity. Moreover, recurrent regime shifts invalidate the assumption of a stationary data-generating process and aggravate the mismatch between historical and future distributions. Data synthesis—defined as the algorithmic generation of statistically

credible yet artificial sequences—has consequently emerged as a pivotal adjunct to prediction workflows: it augments scarce training instances, rebalances the incidence of tail events, and facilitates stress testing by populating hypothetical scenarios. Rather than a mere numerical expedient, synthesis now constitutes an epistemic instrument that enhances both the performance and the trustworthiness of forecasting systems.

Viewed through their algorithmic foundations, extant approaches can be taxonomised into three broad paradigms: classical machine-learning resampling, deep generative modelling, and, most recently, large-language-model-based generation. The first category embeds synthesis within conventional statistical or shallow-learning frameworks; the second leverages variational auto-encoders, generative adversarial networks, and diffusion processes to learn high-dimensional joint distributions; the third repurposes pre-trained language architectures to transfer textual and event semantics into numerical sequences. These families diverge in their representational assumptions, data modalities, and evaluation protocols, and they evolve along a continuum that increasingly foregrounds temporal fidelity, cross-sectional consistency, and semantic interpretability.

Within the machine-learning ambit, early contributions predominantly rely on block bootstrap, Markov-chain Monte Carlo, and sequential Monte Carlo to manufacture alternative realisations of observed series. Brailsford et al. (2016) embed a stationary block bootstrap within GARCH-based value-at-risk estimation and demonstrate a measurable reduction in tail-error rates; Bergmeir et al. (2016) advocate a bagged time-series cross-validation that constructs pseudo-paths under differencing and rolling-window schemes, thereby mitigating over-fitting in non-stationary environments. As the dimensionality of feature spaces expands, synthesis migrates from the response variable to the covariate domain: Liu et al. (2020) apply random-matrix regularisation to factor returns, resample eigenvalues while preserving the Marchenko-Pastur spectrum, and feed the resulting “phantom factors” into a LASSO-autoregression, achieving an 8–12% decline in portfolio forecast error. Although computationally frugal and theoretically grounded, these resampling strategies struggle to reproduce non-linear tail dependence and synchronous jump events, limiting their efficacy in high-frequency or multi-asset contexts.

The advent of deep generative models has prompted a methodological reorientation toward variational auto-encoders, generative adversarial networks, and, more recently, diffusion processes. VAEs endow financial sequences with a low-dimensional latent manifold and optimise the evidence lower bound to ensure distributional consistency. Zhang et al. (2020) decompose intraday returns into trend, jump, and noise latent factors using a β -total-correlation VAE penalised by temporal convolutions; the synthesised five-minute bars pass both Kolmogorov-Smirnov and extreme-jump frequency tests, while an LSTM predictor trained on the augmented set lowers overnight value-at-risk exceedances by 30%. Despite stable training and an explicit likelihood, VAE outputs often suffer from excessive smoothness and under-represent volatility clustering.

GANs, conversely, learn an implicit distribution through adversarial training and are capable of producing visually and statistically sharper sample paths. Koshiyama et al. (2021) employ a Wasserstein-GAN with gradient penalty to generate daily return matrices of the S&P 500 universe while imposing spectral and copula matching constraints; the synthetic "ghost market" is embedded within a deep-reinforcement-learning trading loop and raises out-of-sample Sharpe ratios by 18%. Zhou et al. (2022) extend TimeGAN to the frequency domain, replacing the standard recurrent generator with a bidirectional GRU and incorporating amplitude-phase separation losses, thereby better capturing periodic and chaotic dynamics. GAN training, however, is notoriously unstable; mode collapse and gradient explosions are exacerbated by price jumps, and the absence of a tractable likelihood complicates quantitative assessment, necessitating kernel- or maximum-mean-discrepancy-based diagnostics.

Diffusion models, rooted in stochastic differential equations, iteratively remove noise from a prior distribution and have recently been adapted to financial time series. Liu et al. (2023) propose Score-SDE-FX, which models foreign-exchange tick data as a jump-diffusion process and learns the score function with a time-varying volatility network; macro-announcement conditioning is injected during reverse sampling to enrich jump tail mass. The synthesised 24-hour order flow matches empirical trade intensity and bid-ask spread distributions, while a Transformer predictor trained on the augmented data improves the R^2 of next-day EUR/USD volatility forecasts by 9.4%. Diffusion frameworks yield high-fidelity samples, admit likelihood evaluation, and accommodate physical constraints, yet their multi-step sampling incurs substantial computational cost, impeding deployment in latency-sensitive high-frequency applications.

Parallel to these developments, pre-trained large language models have been harnessed for cross-modal temporal synthesis. Time-LLM (Jin et al., 2023) introduces a "patch-as-prefix" protocol that converts numerical chunks into word embeddings and conditions a frozen LLM through natural-language task descriptions; with only 5% labelled data, the framework lowers MAE on daily S&P 500 forecasts by 14% relative to PatchTST. FinGPT (Yang et al., 2023) further concatenates news releases, central-bank statements, and price ticks into a hybrid text-numerical corpus and fine-tunes an instruction-following LLM to emit both next-day returns and explanatory rationales, thereby integrating prediction and interpretability. LLM-based approaches inherit rich world knowledge and offer human-readable outputs, but their numerical precision is bounded by tokenisation granularity, and context-length limitations complicate high-dimensional cross-asset modelling.

Taken together, the literature reveals an evolutionary trajectory that ascends from distributional moment matching to temporal fidelity and, most recently, to semantic alignment. Concomitantly, evaluation philosophies have expanded beyond single-point error metrics toward a four-tier assessment protocol: (i) distributional tier—marginal densities and cross-asset correlations; (ii) pathwise tier—autocorrelation, volatility clustering, and jump intensity; (iii) factor tier—coverage of asset-pricing factors such as size, value, and momentum;

and (iv) strategy tier—back-test performance, risk-adjusted returns, and drawdown coherence of portfolios trained on synthetic data. This multidimensional benchmarking framework elevates synthetic data from a mere augmentation device to a parallel "virtual market" infrastructure against which models can be stress-tested.

Looking forward, several intertwined research directions promise to refine the synergy between data synthesis and financial forecasting. First, physics-informed generative modelling—embedding stochastic-volatility or jump-diffusion priors within diffusion or LLM architectures—offers a white-box constraint that can enhance extrapolation reliability. Second, multimodal joint synthesis that simultaneously generates order-flow, news sentiment, and options-implied surfaces will enable coherent scenario generation across market segments. Third, online adaptation via streaming variational Bayes or reinforcement learning will allow generators to evolve in real time with regime shifts, mitigating distribution-drift-induced forecast failures. Finally, privacy-preserving techniques such as federated synthesis and differential privacy will facilitate cross-institutional collaboration without compromising proprietary information, aligning technical innovation with regulatory desiderata. As generation methodologies, evaluation frameworks, and compliance protocols mature in concert, data synthesis is poised to become a foundational layer of next-generation financial forecasting systems, providing asset managers, risk officers, and regulators with high-fidelity, interpretable, and scalable data ecosystems.

3 Financial Time Series Data Augmentation

3.1 Task Definition

Financial Time-Series Forecasting (FTSF) is the quantitative endeavour of predicting future values or distributions of financial variables by exploiting the information embedded in their historical realisations. Formally, let

$$\mathcal{X} = \{\mathbf{x}_t \in \mathbb{R}^F\}_{t=1}^T \quad (1)$$

denote a multivariate trajectory of length T with F features (prices, volume, macro-indicators, sentiment scores, etc.). A forecasting model is a mapping:

$$f_{\theta} : \underbrace{\mathbf{x}_{t-L+1:t}}_{\text{look-back } L} \longrightarrow \underbrace{\hat{\mathbf{y}}_{t+H}}_{\text{forecast } H}, \quad (2)$$

where L is the context window and H the prediction horizon. Targets $\hat{\mathbf{y}}$ can be chosen according to the downstream task:

1. **Point forecast:** $\hat{y}_{t+h} \in \mathbb{R}$ (return, volatility, yield).
2. **Interval forecast:** $[\hat{y}_{t+h}^{(\alpha/2)}, \hat{y}_{t+h}^{(1-\alpha/2)}]$ for risk limits.
3. **Distributional forecast:** $P(r_{t+h} \mid \mathbf{x}_{t-L+1:t})$ for full stochastic scenario generation.
4. **Event sequence:** $\{l_{t+1}, \dots, l_{t+H}\}$ with $l \in \{\text{INC, DEC}\}$ for directional path signatures.

3.2 Metrics

Kolmogorov–Smirnov distance (KS). The KS statistic captures the largest vertical gap between the empirical cumulative distribution functions of real and synthetic returns. A small value guarantees that no region of the return spectrum—especially the tails—is systematically over- or under-represented, which is crucial for risk applications that depend on accurate tail mass.

Access:

$$KS = \sup_{r \in \mathbb{R}} |F_{\text{real}}(r) - F_{\text{syn}}(r)|,$$

implemented in Python via `scipy.stats.ks_2samp`.

Wasserstein-1 distance (W_1). W_1 measures the minimum “earth-mover” cost required to transport probability mass from the synthetic distribution into the real one. Because it is sensitive to the geometry of the underlying space, it penalises distant mismatches more heavily than KS and is widely preferred when the shape of the distribution (tails, modes, spread) carries economic meaning.

Access:

$$W_1 = \int_0^1 |F_{\text{real}}^{-1}(u) - F_{\text{syn}}^{-1}(u)| du \approx \frac{1}{n} \sum_{i=1}^n |x_{(i)} - y_{(i)}|$$

(order statistics), available through `scipy.stats.wasserstein_distance`.

Maximum Mean Discrepancy (MMD). MMD embeds each distribution into a reproducing-kernel Hilbert space and compares their mean embeddings. It is a distribution-free test that captures both marginal and joint discrepancies without explicit density estimation, making it attractive for high-dimensional or heavy-tailed financial data.

Access:

$$\text{MMD}^2 = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n k(x_i, x_j) - \frac{2}{n^2} \sum_{i=1}^n \sum_{j=1}^n k(x_i, y_j) + \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n k(y_i, y_j) \quad \text{RMSE} = \sqrt{\frac{1}{M} \sum_{m=1}^M (\sigma_m - \sigma_{\text{target}})^2}$$

with kernel

$$k(a, b) = \exp\left(-\frac{\|a - b\|^2}{2\sigma^2}\right);$$

ready-made in `torchmetrics.functional.maximum_mean_discrepancy`, coded with `numpy.sqrt` and `numpy.square`.

Autocorrelation Error (ACF-E). Financial returns are uncorrelated, but their squares exhibit persistence (volatility clustering). ACF-E quantifies how closely the synthetic series reproduces these temporal dependencies up to a chosen lag L . A small ℓ^2 -error indicates that the synthetic data inherit the same momentum and mean-reversion properties as the real market.

Access:

$$\text{ACF-E} = \|\rho_{\text{real}}(1:L) - \rho_{\text{syn}}(1:L)\|_2$$

where

$$\rho(h) = \frac{\text{Cov}[x_t, x_{t-h}]}{\text{Var}[x_t]};$$

computed with `statsmodels.tsa.acf`.

Ljung–Box $Q(20)$ on squared returns. The Ljung–Box statistic tests the joint significance of autocorrelations in squared returns up to lag 20; a small difference in p-values between real and synthetic data indicates that volatility-clustering behaviour is preserved.

Access:

$$Q = n(n+2) \sum_{h=1}^{20} \frac{\rho_{\varepsilon\varepsilon}^2(h)}{n-h} \quad \text{with} \quad \varepsilon_t = (x_t - \mu)^2;$$

p-values are obtained from `statsmodels.stats.diagnostic.acorr_ljungbox`.

Target-Return Hit-Rate. This metric measures the fraction of synthetic paths whose annualised mean return lands within a user-specified band around the target. It is essential for scenario-based stress testing or strategy back-testing, where the generator must reproduce particular drift conditions.

Access:

$$\text{HR} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}(|\mu_m - \mu_{\text{target}}| \leq b)$$

where

$$\mu_m = \frac{252}{T} \sum_{t=1}^T r_{t,m};$$

implemented with `numpy.mean` and `numpy.abs`.

Realised-Volatility RMSE. RMSE quantifies the generator’s accuracy in reproducing a pre-defined annualised volatility level. A low value ensures that the synthetic environment exhibits the same risk magnitude as the reference, which is critical for risk-management simulations.

Access:

$$\text{RMSE} = \sqrt{\frac{1}{M} \sum_{m=1}^M (\sigma_m - \sigma_{\text{target}})^2}$$

with

$$\sigma_m = \sqrt{\frac{252 \sum_{t=1}^T (r_{t,m} - \mu_m)^2}{T-1}};$$

coded with `numpy.sqrt` and `numpy.square`.

Predictive-Score (PS). PS trains an LSTM on synthetic data and evaluates its MAE on real future returns. A low MAE indicates that the augmented set carries genuine predictive information, validating the utility of the generator beyond distributional fit.

Access:

$$\text{PS} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

where

$$\hat{y}_i = \text{LSTM}_{\theta}(x_{\text{syn}}; x_{\text{val}})[i];$$

evaluated with `torch.nn.L1Loss` or `sklearn.metrics.mean_absolute_error`.

Portfolio Sharpe Ratio. The Sharpe ratio compares the risk-adjusted performance of an RL agent trained within the

synthetic market and then deployed on real data. A higher Sharpe ratio indicates that the generator produces actionable and realistic scenarios.

Access:

$$SR = \frac{\bar{R}_p - R_f}{\sigma_p}$$

where $\bar{R}_p$ and σ_p are computed from the strategy's daily P&L; available through `pyfolio.timeseries.sharpe_ratio`.

99% VaR Coverage Error. This error measures how closely the empirical exceedance rate of the synthetic 99% VaR matches the theoretical 1%. A small absolute deviation is essential for regulatory validation and ensures reliable tail-risk estimates.

Access:

$$Err = \left| \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{r_i < \text{VaR}_{0.99}\} - 0.01 \right|$$

where

$$\text{VaR}_{0.99} = \text{quantile}(\text{synthetic_returns}, 0.99);$$

implemented with `numpy.percentile` and `numpy.mean`.

Membership-Inference AUC. AUC quantifies the privacy risk by measuring how well a discriminator can distinguish real from synthetic records. An AUC close to 0.5 implies strong privacy, indicating that the synthetic data does not leak individual information.

Access:

$$\text{AUC} = \text{roc_auc_score}(y_{\text{true}} = \text{labels}, y_{\text{score}} = \text{probs})$$

with `labels` $\in \{0 = \text{syn}, 1 = \text{real}\}$ and `probs` the discriminator's predicted probabilities; provided by `sklearn.metrics.roc_auc_score`.

ε -Differential Privacy Budget. ε summarises the cumulative privacy loss across all training iterations. A smaller ε means stronger privacy. The budget is computed using the moments accountant for the Gaussian mechanism.

Access:

$$\varepsilon = \sum_t \varepsilon_t \quad \text{with} \quad \varepsilon_t = \sqrt{2 \log \left(\frac{1.25}{\delta} \right)} \cdot \frac{S_t}{\sigma_t};$$

computed via `tensorflow.privacy.privacy.analysis.compute_epsilon_via_random_matrix_theory` and resamples or an equivalent moments-accountant implementation.

3.3 Works Review

Financial Time Series Data Augmentation (FSTDA) refers to the systematic generation of statistically credible and temporally coherent synthetic sequences that enlarge or enrich the empirical sample space on which forecasting models are trained. Rather than merely inflating the number of observations, FSTDA seeks to restore population heterogeneity, to rebalance extremely under-represented tail episodes, and to supply controlled counterfactual regimes so that predictive algorithms can learn robust, generalisable decision rules.

Operationally, the process is framed as a conditional synthesis task: given a historical panel of prices, returns, order flows or derived indicators, one estimates a probabilistic generative mechanism capable of emitting new trajectories whose marginal, joint and dynamic properties remain analytically or behaviourally indistinguishable from those of the genuine market. Once the generator is validated through stylised-fact diagnostics and downstream-task utility tests, its output is pooled with the original records or used to create synthetic-only training folds, thereby expanding the effective sample size, improving generalisation error, and mitigating over-fitting without breaching privacy or regulatory constraints.

The methodological landscape of FSTDA can be partitioned into four broad families whose algorithmic cores differ in representational assumptions, optimisation objectives, and inductive biases. Machine-learning-based approaches primarily rely on classical resampling, perturbation, or latent-factor manipulation; GAN-based paradigms adversarially learn an implicit distribution of sequences; diffusion-based frameworks iteratively denoise random fields while respecting temporal constraints; and a residual category gathers emerging strategies such as neural ordinary differential equations, score-based dynamics, quantum-enhanced sampling, and large-language-model-driven scenario scripting. Each lineage has evolved through successive architectural refinements, regularisation inventions and hybridisation with domain-specific priors, producing a rich toolkit that can be matched to the peculiarities of financial data—non-stationarity, heavy tails, asynchronous quoting, micro-structural noise and pronounced regime shifts.

Machine-learning-based methods constitute the earliest yet still widely deployed stratum of FSTDA. Their underlying philosophy is to manipulate observed realisations without explicitly modelling a full joint density. Block-bootstrap procedures, for instance, retain temporal dependence by resampling contiguous blocks whose length is calibrated to the decay of autocorrelation or conditional heteroskedasticity; stationary bootstrap and tapered-block variants further randomise block size or apply kernel weighting to mitigate edge effects. Markov-chain Monte Carlo (MCMC) techniques draw alternative paths from a recursively estimated transition kernel, while sequential Monte Carlo samplers propagate particles through a sequence of tempered distributions to explore multimodal regions of the state space. A higher-dimensional extension factorises the panel into systematic and idiosyncratic components via random-matrix theory and resamples eigenvalues to preserve the Marchenko-Pastur spectrum, a tactic shown by Liu et al. (2020) to reduce portfolio forecast error by more than eight per cent. Although these approaches are computationally frugal and theoretically transparent, they struggle to reproduce non-linear cross-asset dependence, synchronous jumps or extreme comovements that characterise modern electronic markets.

Generative adversarial networks inaugurated the second generation of FSTDA by casting synthesis as a two-player game whose equilibrium implicitly parameterises the data manifold. Early financial implementations adopted the original GAN discriminator-generator duo, yet mode col-

lapse and vanishing gradients quickly surfaced in the presence of fat-tailed returns. Wasserstein-GAN with gradient penalty (WGAN-GP) stabilised training through a Lipschitz-continuous critic and delivered earth-mover distances that meaningfully reflected distributional shifts; Koshiyama et al. (2021) coupled WGAN-GP with spectral and copula-matching constraints to replicate the cross-correlation structure of S&P 500 constituents and embedded the resulting “ghost market” inside a deep-reinforcement-learning trading loop, raising out-of-sample Sharpe ratios by eighteen per cent. Temporal convolutional generators soon replaced recurrent adversaries, yielding TimeGAN and its financial derivative FinGAN whose exponentially dilated kernels captured long-range dependence without back-propagation-through-time. More recently, conditional architectures have injected macroeconomic indicators, news sentiment or order-book snapshots as auxiliary inputs, enabling scenario-specific generation that supports counterfactual stress testing. Despite these advances, GAN-based FSTDA still contends with hyper-parameter sensitivity, covert memorisation of training samples and the absence of a tractable likelihood, complicating quantitative assessment of synthetic quality.

Diffusion probabilistic models constitute the third and currently most rapidly expanding pillar. They frame synthesis as a reverse-time stochastic differential equation that progressively denoises a Gaussian prior into a data-consistent sample. Score-based extensions parameterise the drift term with a neural network trained to estimate the Stein score, while variance-preserving schedules ensure that the forward noising process admits a closed-form transition kernel. Liu et al. (2023) embed a jump-diffusion prior within Score-SDE-FX to model high-frequency foreign-exchange ticks; macro-announcement conditioning is injected during reverse sampling to enrich jump-tail mass, and a Transformer predictor trained on the augmented data improves next-day volatility R^2 by almost ten per cent. Latent-space diffusions further compress the panel into a low-dimensional manifold before denoising, reducing computational complexity and facilitating the incorporation of economic constraints such as no-arbitrage bounds or volatility-surface consistency. Transformer-based denoisers equipped with cross-attention blocks now accept textual scenario descriptions—“Fed hawkish pivot” or “oil-supply shock”—and translate them into coherent price trajectories, thereby bridging natural-language macro reasoning with quantitative path generation. Although diffusion approaches yield high-fidelity samples and admit likelihood evaluation, their multi-step sampling incurs substantial latency, and careful calibration is required to prevent over-smoothing of extreme events.

Beyond the three dominant paradigms, a heterogeneous collection of emerging techniques is beginning to reshape the FSTDA horizon. Neural ordinary differential equations parameterize instantaneous drift and diffusion functions with deep networks, allowing irregularly sampled and asynchronous quotes to be modeled in continuous time. When coupled with control theory, they can steer synthetic paths toward user-specified terminal statistics while respecting market microstructure. Quantum-enhanced generators exploit variational quantum circuits to sample from expo-

nentially large state spaces; although presently confined to simulators, early work by Yang et al. (2023) shows that a five-qubit Wasserstein QGAN can reproduce the heavy-tail property of S&P 500 returns with a Wasserstein distance below 0.001. Large-language-model-driven augmentation treats numerical sequences as foreign-language corpora: Time-LLM repurposes a frozen Llama backbone via patch-as-prefix reprogramming, while FinGPT concatenates news releases with price ticks and instruction-tunes the ensemble to emit both next-day returns and explanatory rationales. Such cross-modal frameworks promise to integrate qualitative event semantics with quantitative trajectory generation, but they remain vulnerable to tokenisation granularity limits and context-length bottlenecks when scaling to high-dimensional cross-asset universes.

The latest wave of technological advancements is characterised by hybridisation, controllability, and evaluation rigour. Researchers increasingly fuse diffusion backbones with adversarial critics to harness the former’s stable likelihood and the latter’s sharp sample quality; conversely, GAN generators are being equipped with diffusion-based discriminators that score sequences via denoising score matching, mitigating mode collapse without sacrificing speed. Physics-informed architectures embed no-arbitrage drift, volatility-surface consistency or central-bank reaction functions directly into the score network, ensuring that synthetic scenarios respect economic priors. Adaptive privacy budgets allocate tighter ε to borrower-level microdata and looser ε to aggregate market indices, enabling federated institutions to pool information while remaining compliant with evolving data-protection statutes. Meanwhile, the evaluation landscape is maturing into a four-tier protocol that scrutinises distributional fit, pathwise dynamics, factor exposure, and downstream strategy performance, thereby elevating FSTDA from an augmentation trick to a credible virtual-market infrastructure. Looking forward, the convergence of streaming variational Bayes, reinforcement-learning-guided exploration, and causal-counterfactual conditioning is poised to deliver self-evolving simulators that continuously adapt to regime shifts, supply personalised stress scenarios, and provide audit-ready explanations for every synthetic path they emit.

4 Imputation of Financial Time Series Data

4.1 Task Definition

Financial time series imputation aims to recover missing values in multi-dimensional financial data while preserving temporal dynamics, cross-sectional relationships, and economic interpretability. Formally, let $\mathcal{X} \in \mathbb{R}^{T \times N \times D}$ represent a financial time series tensor where T denotes the time dimension (e.g., trading days, months, quarters), N the entity dimension (e.g., firms, bonds, analysts), and D the feature dimension (e.g., financial ratios, stock prices, economic indicators). Define a binary missingness indicator tensor $\mathcal{M} \in \{0, 1\}^{T \times N \times D}$ where $\mathcal{M}_{t,n,d} = 1$ if entry $\mathcal{X}_{t,n,d}$ is missing and 0 otherwise. The observed entries are indexed by set $\Omega = \{(t, n, d) : \mathcal{M}_{t,n,d} = 0\}$. The imputation task seeks to construct an estimator $\hat{\mathcal{X}}$ that minimizes reconstruction error on observed entries while accurately pre-

dicting missing values and preserving temporal dependencies $f(\mathcal{X}_t) = g(\mathcal{X}_{t-1}, \mathcal{X}_{t-2}, \dots, \mathcal{X}_{t-\tau})$ and cross-entity relationships $\mathcal{X}_{t,n,d} \sim h(\mathcal{X}_{t,m,d})$ for related entities n, m . This leads to the general optimization problem:

$$\min_{\theta} \mathcal{L}_{\text{rec}}(\theta) + \lambda \mathcal{R}(\theta)$$

where $\mathcal{L}_{\text{rec}}(\theta) = \sum_{(t,n,d) \in \Omega} \ell(\mathcal{X}_{t,n,d}, f_{\theta}(\mathcal{X}_{\text{obs}}))$ represents the reconstruction loss, $\mathcal{R}(\theta)$ a regularization term, and f_{θ} the imputation function.

4.2 Evaluation Metrics

The literature employs diverse metrics to assess imputation quality across statistical accuracy, economic significance, and model robustness. Statistical accuracy is primarily measured through Root Mean Squared Error (RMSE), $\text{RMSE} = \sqrt{\frac{1}{|\Omega_{\text{test}}|} \sum_{(i,j,k) \in \Omega_{\text{test}}} (\mathcal{X}_{ijk} - \hat{\mathcal{X}}_{ijk})^2}$, Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). Relative performance is often assessed via Relative RMSE (RRMSE) against benchmarks such as last-observation-carried-forward. For financial prediction tasks, Type II Error (false negative rate) and Area Under ROC Curve (AUC) are critical, as missing a bankrupt firm leads to bad loans. Directional accuracy measures the percentage of correctly predicted turning points, while tracking error evaluates performance in index tracking applications. Economic significance is assessed through portfolio returns from long-short strategies constructed using imputed data, statistical similarity between imputed and original distributions, and privacy risk via membership inference attacks. Model quality metrics include F1-score for anomaly detection, Semantic Consistency Score (SCS) for semantic correction, and parameter efficiency ratios.

4.3 Taxonomy of Methods

The Chow-Lin method with LASSO integration represents a classical econometric approach for temporal disaggregation. It estimates monthly values from quarterly data using auxiliary indicators while selecting relevant predictors via LASSO regularization. The model formulates $y_t = \beta'x_t + u_t$ where u_t follows an AR(1) process, with constraints ensuring quarterly averages of monthly imputations match observed values. This method excels in the Swiss business cycle context, achieving superior directional accuracy and external validation against independent economic indicators.

Kernel Ridge Regression for discount curve estimation operates in Reproducing Kernel Hilbert Space to estimate the discount function $g(x)$ from noisy bond prices. It minimizes a weighted least-squares objective $\sum_{i=1}^M \omega_i (P_i - \sum_{j=1}^N C_{ij} g(x_j))^2 + \lambda \|g\|_{\alpha, \delta}^2$, where the kernel $k(x, y)$ incorporates exponential weighting $e^{\alpha x}$ for long-maturity regularization. Hyperparameters are selected via cross-validation, and the representer theorem yields a closed-form solution $\hat{g}(x) = 1 + \sum_{j=1}^N k(x, x_j) \beta_j$. This method generalizes the Smith-Wilson approach and outperforms regulatory benchmarks in the Swiss government bond market.

K-Nearest Neighbors, MICE, and Random Forest imputation leverage supervised learning principles. KNN estimates

missing values as weighted averages of k most similar samples based on Euclidean distance, while MICE iteratively imputes each feature using regression on others until convergence. Random Forest imputation trains ensembles of decision trees to predict missing values, effectively capturing non-linear interactions in financial distress prediction datasets. Empirical studies demonstrate that RF imputation achieves the highest AUC and lowest Type II error at high missing rates (40-50%), significantly outperforming deep learning alternatives like DNN imputation.

The BiGRU-BiLSTM fusion architecture addresses temporal interpolation by leveraging bidirectional dependencies. It employs a two-layer Bidirectional GRU followed by a Bidirectional LSTM encoder to extract spatio-temporal features. The network interpolates missing values based on forward and backward context, then uses a GMDH polynomial regression decoder for final prediction. This approach effectively handles multivariate time series with complex missing patterns in stock market forecasting applications.

NMTucker introduces non-linear Matryoshka Tucker decomposition for tensor completion. It replaces the linear Tucker operation with a neural network-based reconstruction $\hat{x}_{i_1 i_2 i_3} = \text{MLP}([U_{i_1, :}, V_{i_2, :}, W_{i_3, :}])$ and implements recursive core decomposition $\mathcal{G}^{(l+1)} = \mathcal{F}^{\sigma}(\mathcal{G}^{(l)}, U^{(l)}, V^{(l)}, W^{(l)})$ with decreasing ranks to prevent overfitting. This method achieves up to 53.91% RMSE reduction on firm fundamentals and EPS forecast datasets while maintaining parameter efficiency through its hierarchical compression.

RegTensor extends this paradigm with fast non-linear coupled tensor completion and orthogonal regularization. It integrates information across multiple sparse tensors by sharing factor matrices, with the objective:

$$\mathcal{L}_{\text{Rec}} = \sum_{\Omega_X} \|X - \sigma([U, V, T])\|^2 + \lambda \sum_{\Omega_Y} \|Y - \sigma([U, V, W])\|^2,$$

where $\sigma(\cdot)$ is often implemented with a MLP. The orthogonal regularization term $\beta \|U^T U \odot (\mathbf{1}\mathbf{1}^T - I)\|_{2,2}$ enforces uncorrelated embeddings, reducing redundancy. Applied to EPS forecasts coupled with firm fundamentals, RegTensor generates economically significant portfolio returns of 0.7155% monthly, outperforming baselines.

LLM-Forest pioneers ensemble learning of LLMs with graph-augmented prompts for tabular imputation. It constructs bipartite graphs $G_j = (N \cup R_j, E_j)$ linking entries to feature values, with edge weights $w_i(a_k^j) = \log(1 + p_i(a_k^j))$. Hierarchical graph merging aggregates information across features, while random walks retrieve diverse neighbors. Each LLM-tree receives a unique few-shot prompt based on retrieved neighbors, and final predictions combine via confidence-weighted voting (High=1.0, Medium=0.6, Low=0.3). This approach achieves state-of-the-art performance across nine real-world datasets without requiring finetuning.

MTabGen adapts diffusion models for mixed-type tabular data. The forward process adds Gaussian noise to numerical features and multinomial noise to categorical features. The reverse denoising employs a transformer encoder-decoder with conditioning attention, where queries derive

from masked features and keys/values from conditioning features. Columnar embeddings unify numerical (linear+ReLU) and categorical representations. Training minimizes combined losses $\mathcal{L}_{\text{num}}$ and $\mathcal{L}_{\text{cat}}$ on mini-batches with dynamic masking. This unified framework handles both imputation and generation, outperforming TabDDPM and TabCSDI in machine learning efficiency and statistical similarity while providing uncertainty quantification through quantile regression.

4.4 Main Problems and Solutions

Extreme Sparsity and High Dimensionality. Financial tensors exhibit more than 95% missingness due to irregular reporting. Tensor factorization methods address this by exploiting low-rank structure. NMTucker’s recursive Matryoshka decomposition compresses the core tensor from $O(R^3)$ to $O(\log R)$ parameters, while RegTensor shares information across coupled tensors to amplify weak signals. These approaches reduce effective dimensionality while preserving predictive capacity.

Non-Linear Temporal Dependencies Financial relationships exhibit regime shifts and volatility clustering. Neural reconstruction via MLPs in NMTucker and RegTensor captures non-linear interactions among latent factors. BiGRU-BiLSTM architectures leverage bidirectional temporal context, while diffusion models learn complex data distributions through gradual denoising processes. These non-linear formulations significantly outperform linear alternatives in capturing financial dynamics.

Complex Missingness Patterns Missingness in finance is often MNAR and correlated with economic conditions. LLM-Forest uses graph-based retrieval to identify neighbors with similar missingness patterns, treating the mask as an explicit conditioning variable. Kernel Ridge Regression employs duration-based weights $\omega_i = \frac{1}{M(D_i P_i)^2}$ to transform price errors into yield errors, effectively handling missingness correlated with bond characteristics.

Overfitting in Sparse Settings High-capacity models memorize noise when data is scarce. Orthogonal regularization in RegTensor enforces uncorrelated embeddings via $\beta \|U^T U \odot (\mathbf{1}\mathbf{1}^T - I)\|_{2,2}$. NMTucker’s recursive decomposition acts as a hierarchical bottleneck, while LLM-Forest’s ensemble voting reduces variance. These regularization strategies improve generalization without sacrificing model capacity.

Temporal Misalignment Financial data arrives at varying frequencies. The Chow-Lin method with LASSO disaggregates quarterly data using monthly auxiliary indicators, while RegTensor couples tensors with shared time modes but different granularities. These approaches synchronize heterogeneous temporal structures while preserving economic relationships.

Auxiliary Information Integration Sparse target tensors benefit from denser auxiliary data. RegTensor’s coupled factorization shares factor matrices across related tensors, allowing information flow from firm fundamentals to EPS forecasts. BBGMDH employs multi-task learning with shared encoders for interpolation and prediction. This cross-tensor

information transfer improves imputation accuracy and economic significance.

Temporal Extrapolation vs. Interpolation Imputing beyond observed horizons is fundamentally difficult. KR uses exponential weighting $e^{\alpha x}$ to discourage wild extrapolation, while the tensor factor model suggests re-imputation with a larger rank for long-range accuracy. The Chow-Lin framework cautions against extrapolation beyond 30-40 years without external information, recommending constraint-based anchoring to economic fundamentals.

Mixed Data Types Financial data combines continuous and categorical features. MTabGen employs separate diffusion processes for each type, while RegTensor and NMTucker use unified columnar embeddings. LLM-Forest’s natural language prompting naturally handles heterogeneous features. These strategies maintain statistical properties across data types within a single framework.

Economic Interpretability and Fairness Imputed values must be economically plausible and bias-free. Diffusion models provide uncertainty quantification through confidence bands. The Fair MIP Forest avoids imputation entirely by training decision trees directly on incomplete data with fairness-regularized objectives $\ell(\mathcal{D}) + \lambda \ell_{\text{fair}}(\mathcal{D})$. Chow-Lin validates imputations against external business cycle indicators, ensuring economic relevance.

Computational Scalability Financial datasets often contain millions of observations. Stochastic optimization restricts updates to observed entries, reducing complexity from $O(n^3)$ to $O(|\Omega|)$. LLM-Forest’s graph construction operates in $O(nd)$ versus KNN’s $O(n^2 d)$, while ensemble trees enable parallel computation. These optimizations make advanced methods feasible for production systems.

In conclusion, the financial time series imputation landscape has evolved from simple statistical methods to sophisticated deep learning and generative models, each addressing specific challenges of financial data. Tensor-based methods excel at high-dimensional sparsity, RNNs capture temporal dynamics, LLMs provide semantic understanding, and diffusion models offer uncertainty-aware generation. The optimal approach depends on data characteristics: tensor structure favors NMTucker/RegTensor, pure time series benefit from BiGRU-BiLSTM, and tabular data with semantic noise leverages LLM-Forest. Future directions include real-time streaming adaptation, few-shot domain specialization, and hybrid architectures combining these complementary strengths.

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